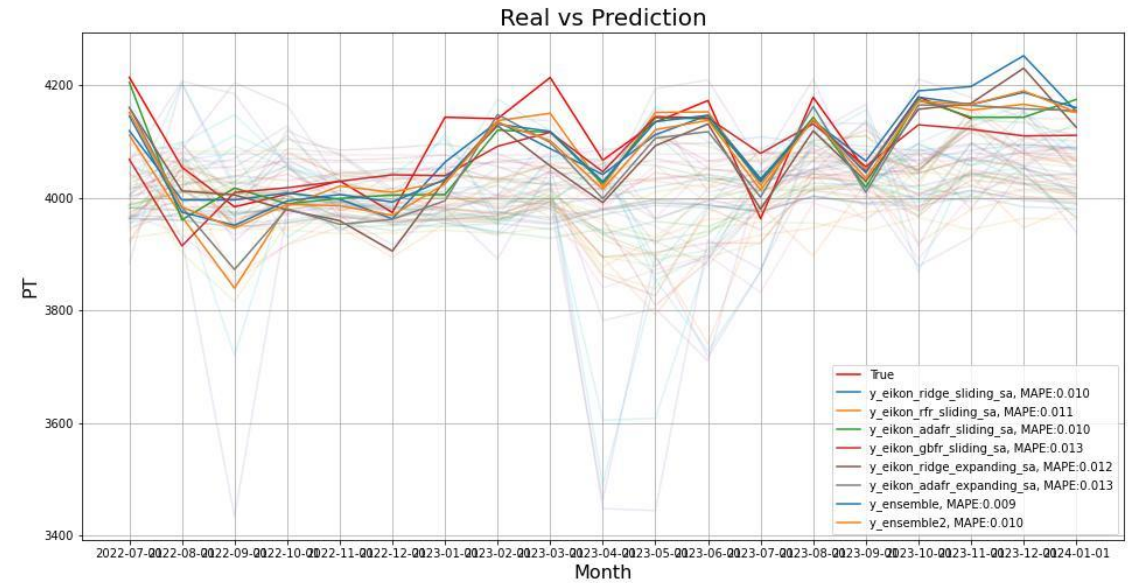
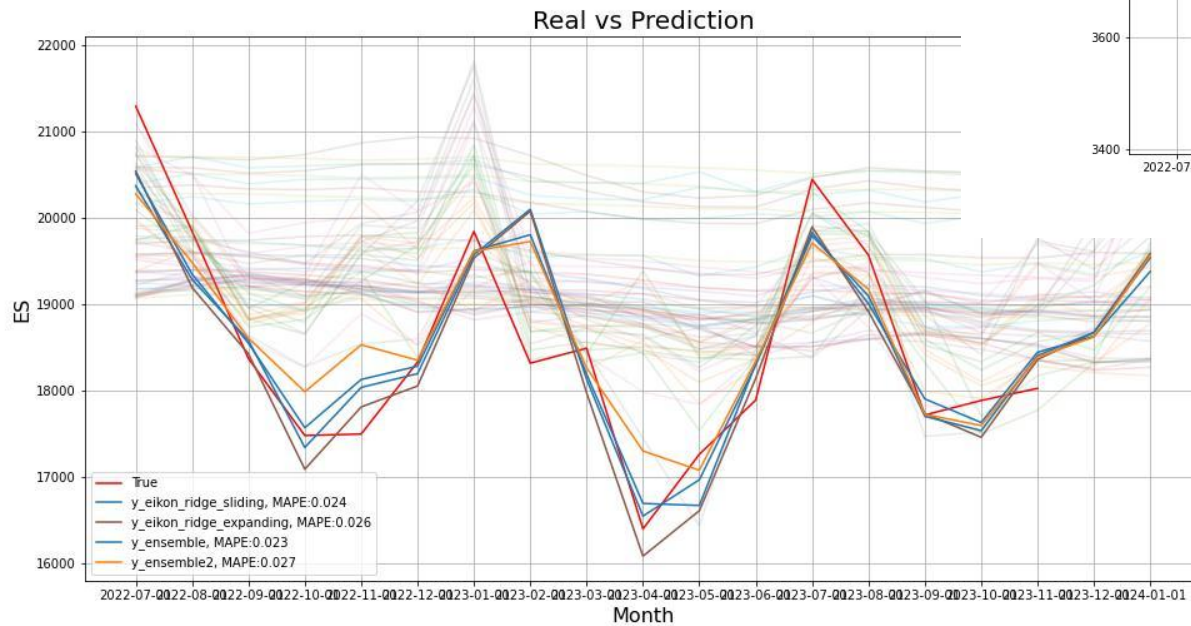


Prelude

- Eurostat Nowcasting competition
- Nowcasting for: PVI, PPI, and tourism
- Optimizing classical ML algorithm
- Limited resources of data



General approaches

Methods

- Smart-up traditional ML algorithms with PSO hyperparameter search, e.g. PSO + AdaBoost
- Use recursive NN, which can handle time series, such as LSTM units, e.g. LSTM unit + Dense layers, Encoder-Decoder architecture
- Gaussian processes regressor for probabilistic interpretation and benchmarking.

Evaluation

- 80%-20% train-test/validation split
- Remove trend, if needed
- Feature extraction and scaling when needed
- Metrics: MAPE and SD
- Method for feature importance

Implementation

- Compact object-oriented library for models, routines
- Nowcasting depending on data arrivals (T+0-T+15)



$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - p_i}{y_i} \right|$$

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Neural network LSTM models

Preprocessing

Make time series stationary: e.g. remove trend

Feature scaling: important for NN models

Model input: fixed length rolling window segments

Architecture

LSTM architecture to featurize time series, e.g.

simple Bidirectional LSTM or encoder-decoder

Additional dense layers with some nonlinear activation, e.g. LeakyRelu

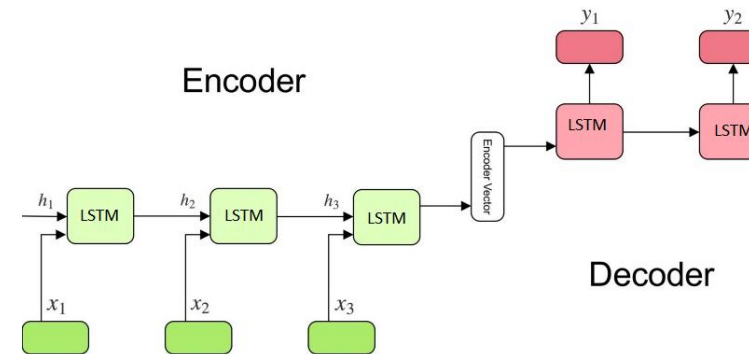
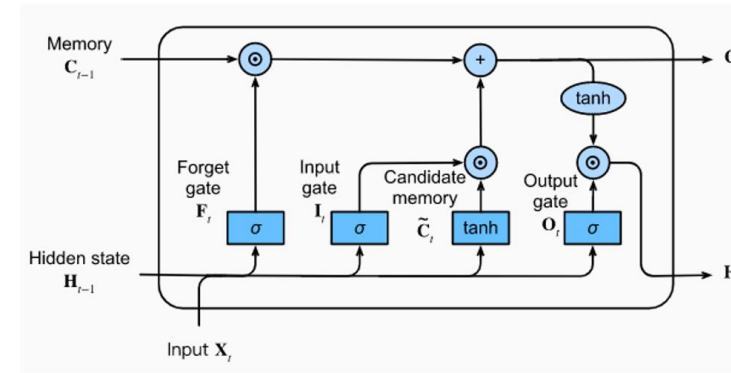
Important hyperparameters

Input sequence length: problem dependent

LSTM output size: 8

Dense layer size: 8

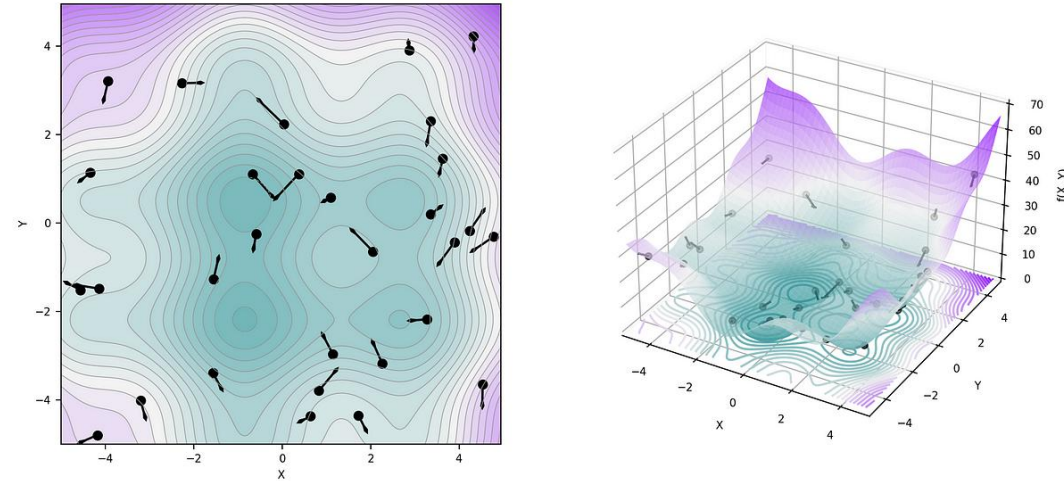
Learning rate: 1e-4



PSO: Particle Swarm Optimization

- Optionally minimal preprocessing, i.e. scaling

particles move depending both on local gradient and global minima of the flock

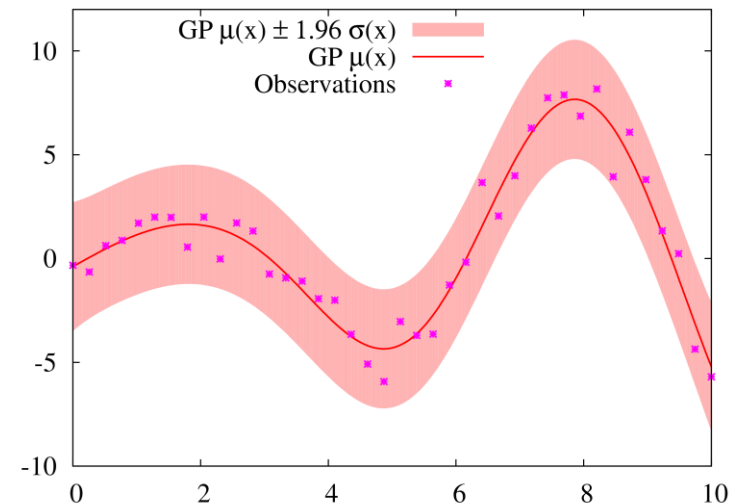


Boosting or even ensemble models



Gaussian processes regression

- No preprocessing needed
- Estimates mean and covariance function of the time series
- Predicts distributions (probabilistic interpretation)



Benchmark results on simulated data.

